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# Design Factors for Computer Games in Public Spaces With a Variable Number of Players

Bachelor Thesis (Proposal)

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# Kurzfassung

Large Public Display Games – Spiele auf großen Projektionsflächen in öffentlichen Räumen – stellen sehr spezifische Anforderungen an das Gamedesign. So müssen diese Spiele sowohl für wenige als auch viele Personen funktionieren, ebenso soll ein leichter Ein- und Ausstieg gewährleistet sein. Diese Bachelorarbeit (Masterarbeit) erarbeitet ein Framework aus Spielmechaniken, die das Prinzip des Smooth Transition Gameplays unterstützen und testet diese anhand eines im Rahmen eines Semesterprojekts entwickelten Prototyps.

# Abstract

Large Public Display Games place several specific design requirements. Such games need to work equally well for just a few or several simultaneous users. Also, entering, leaving, or joining a game in progress should be easily possible without interrupting the game's flow. This bachelor (master) thesis focuses on developing a game mechanics framework that supports the principle of smooth transition gameplay. This framework will then be evaluated utilizing a prototype implemented in a term project.

# Exposé

## 1 Introduction

The Deep Space 8K[1] at the Ars Electronica Center in Linz, with its $16 \times 9$ meter projection surface, including position tracking, offers a unique opportunity to create computer games. These games do not use classic control mechanisms such as a keyboard, mouse, or gamepad; instead, the players themselves "control" the content with their movements. Furthermore, these games take place in a semi-public to public space, making it difficult to determine the target group and the number of people playing. This bachelor thesis (master thesis) illuminates this problem and presents concrete solutions based on an example.

## 2 Theoretical Background and State of the Art

Large public display games (LPD games) are a particular type of computer game displayed on large, publicly visible projection surfaces. Such installations can be found in museums (such as the Ars Electronica Center) or public places. People can usually see these games at any time and actively participate in them. According to [1], this publicity results in three groups of people participating in the game: *Actors* actively participate in the game, *spectators* actively follow it, and *bystanders* are just in the vicinity of the public installation. The goal is for bystanders to become spectators and spectators to become actors, that is, to actively play the game. This process should be as fluid as possible and involve as many people as possible. Such an approach was called *Smooth Transition Gameplay* in [2]. The authors use a concrete application to demonstrate how this transition can be achieved, but it needs to be systematically described which factors are necessary.

The used game mechanics provide a starting point. Following the categorization in [3], mechanics from the areas of space, actions, and rules are particularly suitable. Such mechanics can be used in a corresponding game design so that the requirements mentioned above—easy entry and good scalability with respect to the number of players—can be achieved in an LPD game.

[1] https://ars.electronica.art/center/en/exhibitions/deepspace/

## 3 Research Question

These approaches lead to the following research question for this bachelor thesis (master thesis):

> In a game design for a large public display game, which game mechanics have to be used in which way to design it for a variable number of players and to enable an easy entry for them?

## 4 Methodology

To answer this question, the bachelor thesis (master thesis) will be realized as a combination of literature work and practical or prototypical implementation.

First, the existing literature (extending section 2) will show how the topic of smooth transition gameplay is dealt with from a game design perspective. Common factors such as mechanics will be extracted from this and serve as the basis for a theoretical framework. This framework will contain a list of core mechanics and guidelines for their application so that LPD games allow easy entry and a variable number of players.

The applicability of this framework will be tested by an own LPD game developed during the term project. By asking simple qualitative questions to the players and observing the visitors during several test runs, it will be determined whether smooth transition gameplay could be achieved with the mechanics used.

## 5 Expected Results

As a concrete result, a framework of game mechanics will be created to serve as a basis for the creation of LPD games. It is expected that such concrete mechanics can be found and described.

The tests of the practical implementation of the framework are also expected to be positively evaluated since there are already successful concepts and LPD games that can serve as positive examples.

# Documentation of Generative AI Tool Usage

## Responsible use of the AI tools

| **Statements on responsible use of AI tools** | **Confirmation: Yes** |
|---|---|
| I/we confirm that the outputs generated by AI tools have been verified for accuracy and corrected where required. | ☒ |
| I/we acknowledge that the responsibility for the thesis lies with the author(s), and not with the AI tools or any other person. | ☒ |

## Detailed activities for which the tools were used

| **Activity** | **Description** | **Tools used (if any)** | **Description of the tool usage** (type of usage, affected sections of the paper/thesis, etc.) |
|---|---|---|---|
| **Idea generation & conceptualization** | • Generating ideas, objectives and research questions<br>• Identifying and defining relevant concepts and theoretical models<br>• Development of a conceptual framework for the content | — | — |
| **Literature search and analysis** | • Search for relevant literature<br>• Reviewing potentially relevant literature<br>• Summaries of relevant literature | Perplexity | Search for literature on ... (Chapter 2) |
| **Methodology** | • Search for an appropriate methodology<br>• Development and adaptation of the methodology to the research question(s) | — | — |

| **Activity** | **Description** | **Tools used (if any)** | **Description of the tool usage** (type of usage, affected sections of the paper/thesis, etc.) |
|---|---|---|---|
| **Coding** | • Creation and documentation of code, algorithms, and software<br>• Testing and debugging of existing code, algorithms, software<br>• Understanding and analysis of existing code, algorithms, and software | — | — |
| **Data collection and analysis** | • Collection of primary or secondary data<br>• Qualitative data analysis (including summarizing and coding)<br>• Quantitative data analysis (including statistics)<br>• AI-assisted mathematical, computational, or other formal techniques for modeling, simulation, and analytics | — | — |
| **Interpretation and Validation** | • Interpretation of results<br>• Derivation of implications for research and practice<br>• Verification of the overall replicability/reproducibility of the results and other research outputs | — | — |
| **Structuring and planning the text** | • Structure of the thesis<br>• Structuring of individual sections of the thesis (e.g., bulletpoint outlines per section) | — | — |
| **Generating the text** | • Generation of text for topics in various sections of the thesis (including title and abstract) | — | — |
| **Translating text** | • Translation of text written by the author(s)<br>• Translation of text written by others | — | — |
| **Reviewing & editing the text** | • Critical review, feedback or revision of the content or grammar of the thesis<br>• Proofreading<br>• Rephrasing or paraphrasing text<br>• Shortening / expanding the text | DeepL Write | Proofreading of Chapters 1–5 |
| **Citation Management** | • Creation of the reference list<br>• Formatting of the references | — | — |

### Further use of AI tools:

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Source reference: Gimpel, H., Dilger, P., Lämmermann, L., Urbach, N. (2023). COGNISANCE: Declaration of Generative AI Tool Usage in Higher Education Tests. Research Center for Information Management, May 22, 2023. https://doi.org/10.17605/OSF.IO/VGXF7. Adapted by the University of Applied Sciences Upper Austria. Licensed under CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0/).

# References

## Literature


[1] Matthias Finke et al. “Lessons Learned: Game Design for Large Public Displays”. In: *Proceedings of the 3rd International Conference on Digital Interactive Media in Entertainment and Arts*. DIMEA ’08. New York, NY, USA: ACM, 2008, pp. 26–33. DOI: 10.1145/1413634.1413644 (cit. on p. 1).

[2] Wolfgang Hochleitner et al. “Limelight – Fostering Sociability in a Co-located Game”. In: *Proceedings of the CHI 2013 Workshop on Designing and Evaluating Sociability in Online Video Games*. CHI ’13. Paris, France, 2013, pp. 23–28 (cit. on p. 1).

[3] Jesse Schell. *The Art of Game Design. A Book of Lenses*. 3rd ed. Boca Raton, FL, USA: CRC Press, 2019. DOI: 10.1201/b22101 (cit. on p. 1).